

Unsupervised Discovery of Market Regimes Using Static and Temporal Models

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REPOSITORY: <https://github.com/nickdatak/Unsupervised-Discovery-of-Market-Regimes>

Motivation

Financial markets exhibit distinct periods of behavior characterized by differing levels of risk, volatility, and macroeconomic conditions. These periods - usually referred to as *market regimes* - play a crucial role in portfolio construction, risk management, and macroeconomic analysis in general. Accurately identifying market regimes is a fundamental issue in economics, finance, and applied data science.

Traditional approaches to identify and characterise regimes often rely on predefined economic rules or labels, such as recession indicators, volatility thresholds, etc. While valuable and interpretable, these methods set strong assumptions about regimes and often fail to capture gradual transitions or latent market dynamics. On the other hand, modern machine learning approaches are increasingly applied to financial data, but frequently focus on predictive performance rather than interpretability or stability, limiting their usefulness in understanding underlying market behavior.

Unsupervised learning offers a good alternative by allowing regimes to emerge directly from data without predefined labels. However, the extent to which different unsupervised modeling assumptions affect the stability, interpretability, and economic relevance of discovered regimes remains insufficiently explored in applied settings. In particular, systematic comparisons between static clustering methods (Gaussian Mixture Models, *GMM*) and temporally-aware probabilistic models (Hidden Markov Models, *HMM*) on macro-financial indicators are relatively scarce in applied settings.

This project addresses this gap by comparing different unsupervised regime discovery methods using weekly macro-financial data between 1995 and 2024. By evaluating *Gaussian mixture models* and *hidden Markov models*, the study focuses on *regime stability*, *cross-model agreement*, and *economic interpretability* rather than predictive accuracy.

The goal is to provide a clearer understanding of how modeling choices influence the regimes that emerge and their potential usefulness for economic and financial analysis. In simple words, I want to figure out how different unsupervised modeling decisions change the way markets are partitioned into regimes, and whether those regimes are stable, interpretable, and meaningful for analyzing market risk and behavior.

Research questions

This study aims to address the following research questions:

- How stable are the market regimes identified by different unsupervised learning methods when applied to weekly macro-financial data?
- Does incorporating explicit temporal structure (e.g., Hidden Markov Models) lead to more persistent and interpretable regimes compared to static clustering methods (e.g., Gaussian Mixture Models)?
- To what extent do different unsupervised models agree on the timing and structure of identified market regimes?
- Are the identified regimes associated with distinct economic characteristics, such as differences in volatility, drawdowns, and market stress indicators?
- How sensitive are the identified regimes to modeling choices such as the number of regimes and feature selection?

Scope & Non-Goals

Scope

This study focuses on the unsupervised identification and evaluation of market regimes using weekly macro-financial indicators. The analysis is conducted at the aggregate market level, with the S&P 500 serving as a proxy for overall equity market behavior. The time horizon spans multiple market cycles to capture a range of economic conditions, including periods of expansion, stress, and crisis.

The modeling scope is limited to unsupervised learning methods commonly used for regime discovery, including static clustering approaches (e.g., Gaussian Mixture Models) and temporally-aware probabilistic models (e.g., Hidden Markov Models). All models are evaluated under a consistent experimental framework, using the same data, features, and evaluation criteria.

Evaluation emphasizes regime stability, cross-model agreement, and economic interpretability, rather than predictive accuracy. Regimes are assessed based on their persistence over time and their association with economically meaningful characteristics such as volatility, drawdowns, and market stress indicators.

Non-goals

This project explicitly does not aim to:

- Predict future asset prices or returns
- Develop or evaluate trading strategies
- Generate alpha or optimize investment performance
- Perform supervised learning using predefined regime labels
- Apply deep learning or high-frequency modeling techniques
- Analyze individual securities or firm-level data

These exclusions are intentional and reflect the project's focus on understanding and evaluating unsupervised regime discovery methods, rather than forecasting or decision-making applications.

Data Description

Data Sources

The research uses publicly available macro-financial data sets that ensure transparency and reproducibility. Equity market behavior is represented using the S&P 500 index, while market uncertainty and stress are captured through the CBOE Volatility Index (VIX). Macroeconomic conditions are incorporated via interest rate term structure measures sourced from the Federal Reserve Economic Data (FRED) database.

Specifically, the dataset includes:

- S&P 500 Index (price levels and trading volume), sourced from Yahoo Finance
- CBOE Volatility Index (VIX), sourced from Yahoo Finance
- 10-year minus 2-year U.S. Treasury Yield Spread, sourced from FRED

Optional credit market indicators (e.g., BAA–AAA corporate bond spreads) are considered when data availability and quality permit.

Time Period & Frequency

All series are converted to a weekly frequency, spanning the period 1995–2024, subject to data availability.

Modelling approach

The project applies three complementary unsupervised methods to identify market regimes:

- Gaussian Mixture Model (GMM) – a static clustering method*
- Hidden Markov Model (HMM) – a temporally-aware probabilistic model*

These methods are selected to span a spectrum of temporal assumptions:

- GMM assumes each observation is independent
- HMM explicitly models regime persistence and transition probabilities

All models are applied to the same weekly macro-financial dataset to ensure comparability.

**Utilizing Full Covariance matrices to capture feature correlations, e.g., VIX vs. Yields*

Evaluation strategy

Because the project focuses on unsupervised regime discovery, evaluation emphasizes stability, interpretability, and robustness, rather than predictive accuracy. The key idea: A good regime model identifies persistent, economically meaningful market states that are consistent across methods and parameter choices. Evaluation is therefore divided into four pillars:

- Regime Stability & Quality: Measures whether the identified regimes are persistent and meaningful over time using mean regime duration (average number of consecutive weeks per regime), transition frequency (how many times do regimes switch), and entropy of regime transitions (low entropy signaling more predictable regimes, high entropy signaling otherwise). Our goals are obviously high stability that would reflect the true market structure, rather than useless noise.
- Cross-Model Agreement: Assesses whether different models produce consistent regime structures using ARI (Adjusted Rand Index) and visual alignment of regime sequences (timeline plots). Our aim is high agreement across models that would increase confidence that found regimes are robust and are not simply model artifacts.
- Economic Interpretability: Tests whether regimes correspond to economically meaningful patterns using average and standard deviation of returns per regime, volatility (realized and VIX) per regime, MDD (Maximum drawdowns per regime), and correlation with macro stress indicators (e.g., yield spreads). Our goal is simply to be able to explain model findings and differentiate between the periods of high and low market stress (i.e., we should have an intuitive financial explanation).
- Robustness Check: basically evaluates sensitivity to the model choices. Our results should not change under small model modifications. Besides, to address look-ahead bias, we will perform an expanding window analysis (e.g., fit on 1995–2010, predict 2011) to test if regimes identified in-sample generalize to out-of-sample data.

Summary

Evaluation focuses on:

<i>Pillar</i>	<i>Why does it matter?</i>	<i>How do we measure it?</i>
Stability	Are regimes persistent?	Duration, Transitions, Entropy
Cross-Model Agreement	Are results robust across methods?	ARI, Visual Timelines
Economic Interpretability	Are regimes meaningful?	Volatility, Drawdowns, Stress Correlation
Robustness	Are findings sensitive to choices?	K variation, Features, TCHC parameters, Expanding Window.

*No predictive metrics (accuracy, R^2 , Sharpe) are used, consistent with the unsupervised focus.

Definition of success

The project will be considered successful if identified regimes are stable, interpretable, and consistent across models, and the findings are reproducible and robust to variations in model parameters and feature selection.

Results

1. Regime Characteristics and Economic Interpretation

Across all models, the identified regimes exhibit clear and economically interpretable differences in market behavior. We categorize these regimes into three distinct states: Calm (bull markets), Transitional (uncertainty/correction), and Stressful (crisis/bear markets).

I compiled interpretations of the regimes by 2 models into multiple tables for better understanding:

	VIX - HMM	VIX - GMM
Calm	-0.461469	-0.593623
Transitional	-0.287374	0.617895
Stressful	1.047057	1.491176

	S&P500 Returns - <i>HMM</i>	S&P500 Returns - <i>GMM</i>
Calm	0.076538	0.084711
Transitional	0.016970	-0.113450
Stressful	-0.119910	-0.117930

	Yield Spread - <i>HMM</i>	Yield Spread - <i>GMM</i>
Calm	-0.775677	0.135306
Transitional	0.798230	0.548056
Stressful	-0.483853	1.173439

**I use mean values in all of the tables*

Stress Regimes: Both models identify stress periods characterized by negative equity returns and sharply elevated volatility ($VIX > 1.0$). Notably, the GMM "Stressful" regime exhibits a higher mean VIX (1.49) than the HMM (1.05). This confirms that the static GMM isolates extreme, short-lived volatility spikes, whereas the HMM "smoothes" these spikes into broader, more persistent regime blocks.

Calm Regimes: Both models strongly agree on calm periods, marked by below-average volatility and positive equity drift.

Transitional Regimes: This state captures the "middle ground." The GMM identifies this as a moderately volatile state ($VIX 0.50$) often associated with negative returns, likely capturing corrections that do not evolve into full crashes.

2. Regime Stability and Persistence

A fundamental divergence appears in regime duration. The HMM produces substantially more persistent regimes compared to the GMM, reflecting its explicit modeling of transition probabilities.

- **Average Regime Duration:** The HMM regimes persist for an average of ~50 weeks, aligning with medium-term economic cycles. In contrast, the GMM regimes are short-lived, averaging only ~8.4 weeks.⁺¹
- **Transition Entropy:** The HMM exhibits significantly lower transition entropy (0.07–0.16) compared to the GMM (0.36–0.44), indicating that the HMM offers a more stable and predictable signal for risk management.*

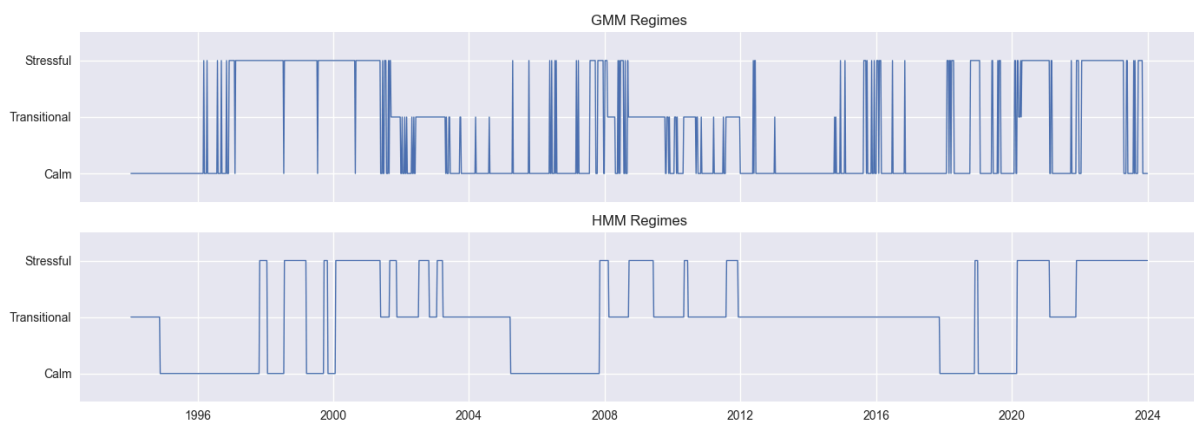
The GMM's tendency to "flicker" between states suggests it treats market data as independent weekly observations, making it sensitive to noise but highly reactive to sudden shocks. The HMM absorbs this noise, trading reactivity for stability.

**Specific entropy figures:*

- GMM transition entropy: 0.36340749, 0.42829457, 0.445967;
- HMM transition entropy: 0.07757004, 0.1643722, 0.08390038;

Here are some visualisations that clearly outline the differences between the 2 models:

Regime timelines:



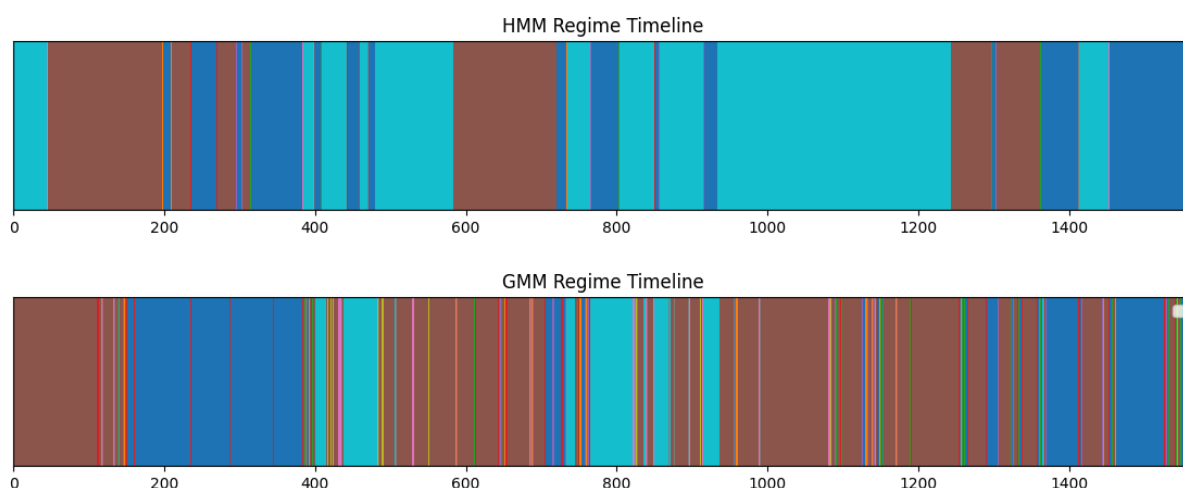
As we can see, the GMM model fluctuates much more often than the HMM model

3. Cross-Model Agreement

Despite differences in modeling assumptions, the GMM and HMM exhibit meaningful agreement in their regime assignments. The Adjusted Rand Index (ARI) indicates non-trivial overlap between the models, suggesting that both capture a shared underlying structure in the data rather than entirely model-specific artifacts. (Adjusted Rand Index (GMM vs HMM): 0.2537944031819488)

To be more specific, we can notice that models agree strongly on stress periods and disagree more on calm and transitional periods. These differences are consistent with the distinct modeling assumptions of static versus temporally-aware methods.

Visual inspection of regime timelines reveals particularly strong alignment during periods of pronounced market stress, such as major financial crises. During calmer market phases, agreement is lower, reflecting greater ambiguity in regime boundaries and sensitivity to modeling choices.



This pattern suggests that market stress regimes are more robust and consistently identifiable across methods, whereas transitional or calm regimes are inherently harder to define precisely.

4. Drawdowns and Risk Characteristics

Maximum drawdown analysis further reinforces the economic interpretability of the identified regimes. Stress regimes across both models are associated with significantly larger drawdowns compared to calm and transitional regimes, confirming their relevance for risk assessment.

Importantly, these drawdown differences emerge without explicitly optimizing for returns or risk metrics, underscoring the ability of unsupervised methods to recover meaningful risk regimes purely from macro-financial structure.

5. Observations on Temporally Constrained Hierarchical Clustering (TCHC)

I initially considered a temporally constrained hierarchical clustering approach to provide a non-probabilistic alternative for regime discovery. However, empirical results showed that TCHC behavior is highly sensitive to the strength of temporal constraints. Weak constraints led to frequent regime switching similar to static clustering, while overly strict constraints resulted in unrealistically long regimes. A moderate temporal window produces regime persistence between GMM and HMM. Despite this, the results remained highly sensitive to linkage and window choices, producing either implausibly short or excessively persistent regimes. Consequently, we exclude this method from the final analysis to prioritize models with more robust and interpretable parameters.

6. Robustness to Modeling Choices

Sensitivity analysis indicates that the qualitative structure of the identified regimes is robust to reasonable variations in model parameters, including the number of regimes and feature selection. While exact regime boundaries and durations vary, the overall economic interpretation and relative ordering of regimes remain stable.

An expanding window analysis further suggests that regimes identified in earlier periods generalize to later data, supporting the stability of the discovered regime structure over time.

7. Summary of Key Findings

Overall, the results demonstrate that:

- Unsupervised models can identify economically meaningful market regimes without predefined labels.
- Incorporating temporal structure (HMM) leads to more persistent and stable regimes.
- Static clustering (GMM) captures sharper distinctions but at the cost of increased regime switching.
- Market stress regimes are the most robust and consistently identified across models.
- Modeling choices materially affect regime structure, highlighting the importance of methodological transparency in regime analysis.

These findings support the use of unsupervised regime discovery as a tool for exploratory financial analysis and risk characterization, while emphasizing that regime definitions are inherently model-dependent rather than objective ground truth.

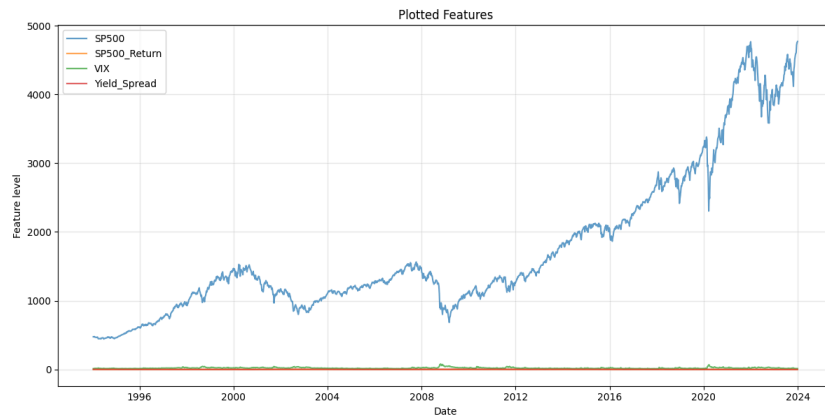
8. Limitations

While this study demonstrates the utility of unsupervised learning for market regime discovery, several limitations must be acknowledged:

- Sensitivity to Parameter Selection (k): The number of identified regimes (k) was fixed at three (Calm, Transitional, Stressful) for comparability. However, unsupervised models are sensitive to this choice, and a different k could reveal more granular sub-regimes or merge distinct economic states.
- Weekly Data Granularity: The analysis uses a weekly frequency to capture macro-financial trends. This frequency may overlook intra-week volatility or rapid "flash" regime shifts that occur at a daily or high-frequency level.
- Proxy Limitations: The S&P 500 is used as a proxy for the entire equity market. While representative of large-cap US equities, the identified regimes may not fully capture dynamics in small-cap stocks, international markets, or other asset classes like fixed income.
- Temporal Model Assumptions: While the HMM accounts for temporal dependence, it assumes the transition matrix is stationary over the entire 1995–2024 period. In reality, the "rules" of market transitions may shift over decades due to structural changes in the economy.

Additional visualisations:

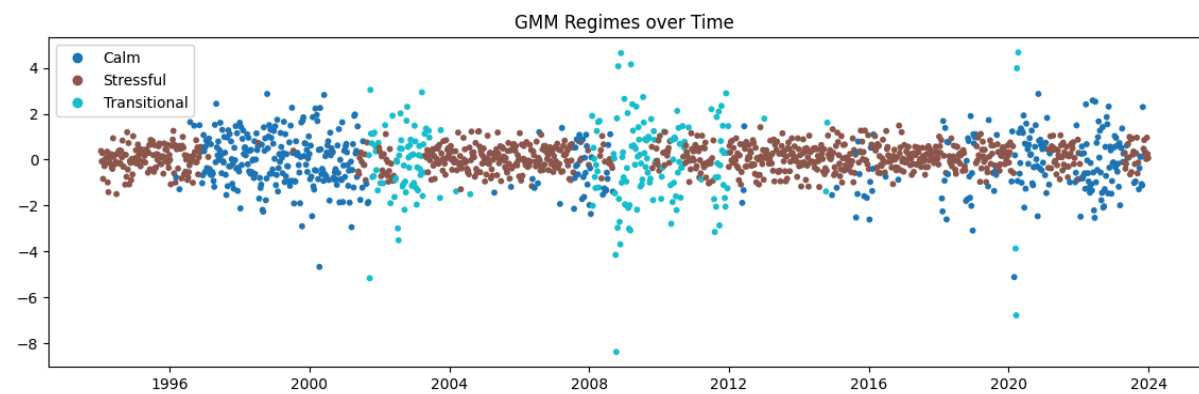
Market features:



Standardised market features:



GMM scatter plot:



HMM scatter plot:

